

LEO Satellite Network Resilience Analysis: A Focus on Critical Satellites

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ABSTRACT

The rapid development of Low Earth Orbit (LEO) satellite technology is revolutionizing global internet infrastructure. However, limited work has been conducted on network resilience for the next generation of satellite networks based on Inter-Satellite Links (ISLs). This study investigates LEO satellite network resilience under possible physical damage to ISLs-based satellite constellations using Starlink Shell 1 and Kuiper Shell 3 as a case study. We demonstrate the geographical distribution of critical satellites and reveal that the malfunction of these critical satellite results in a significant increase in the workload of the remaining satellites, thereby notably reducing the overall communication performance of the satellite network.

CCS CONCEPTS

• **Networks** → **Network simulations**; **Network performance analysis**; *Network measurement*;

KEYWORDS

LEO Satellites, Simulation, Network Resilience

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1 INTRODUCTION

LEO satellites, addressing the difficulty of reaching remote areas with ground-based networks, are emerging as a promising internet backbone with higher coverage and reduced latency [2]. Since the first LEO satellite launch in 2019, Starlink has deployed over 6,000 satellites, exceeding 3 million users [34], and is projected to generate 6.6 billion dollars in revenue in 2024 [16]. Other companies, including Kuiper [1], OneWeb [26], and Telesat [35], are also developing their satellite constellations. The next generation of LEO constellation internet design is progressing from theory to active, promising construction.

The LEO satellite network offers more than reduced latency due to its proximity to Earth. With over ten thousand satellites capable of high-speed, direct inter-satellite communication, it has the potential to handle end-to-end internet traffic entirely in space using inter-satellite links (ISLs). Starlink's satellites, equipped with advanced optical laser technologies, exemplify this capability with data transfer rates reaching up to 80 Gbps [6, 7, 12, 13, 34, 37]. The +Grid topology is a well-studied configuration for ISLs, where satellites connect with their immediate neighbors in the same and adjacent orbits. This structured approach optimizes network resilience and minimizes latency variability [2, 3, 14, 32, 39]. Unlike bent-pipe routing strategies reliant on ground stations [21], +Grid ISL enables direct, light-speed communication between satellites, eliminating the need for ground stations. Starlink has already activated ISLs in certain regions [10]. Furthermore, Starlink's next-generation V2 satellites, in collaboration with T-Mobile, aim to provide direct cellular device services via ISLs [30]. Elon Musk envisions Starlink becoming a major player in global backhaul communications and local internet traffic, potentially cutting global real-time network latency by up to 50% under optimal conditions [9]. This vision represents a transformative shift towards a next-generation space-based internet backbone, characterized by high throughput, extensive global coverage, and significant latency reduction capabilities.

Despite the notable advantages of LEO satellite networks over traditional terrestrial infrastructure, they face significant challenges. The network's vulnerability and instability

primarily stem from its critical infrastructure—the satellites themselves. In constellations employing ISL routing strategies, each satellite in various orbits acts as a crucial node in the +Grid network responsible for routing traffic. The failure or disconnection of any satellite along the routing path can trigger cascading effects on overall network performance. Such issues range from momentary packet loss and increased re-transmission to severe degradation or complete failure of ISL satellite network connectivity. Moreover, the replacement and commissioning of a new satellite after failure or damage often cannot be swiftly accomplished. This is particularly problematic for services and communication infrastructure heavily reliant on satellite networks, such as disaster relief operations [5], earth surveillance systems [36], and military communications [31].

At an altitude of 500 km above Earth, satellites traveling at 27,000 km/h face formidable challenges in the harsh conditions of space. Currently, satellites encounter several significant threats: **(1) Malfunctions and Scrapping.** Initial batches of Starlink satellites experienced a 5% failure rate in orbit [25]. SpaceX plans to deorbit approximately 100 older Starlink satellites in 2024 due to design flaws [11]. Non-functioning satellites remain in orbit for up to five years before atmospheric re-entry, complicating replacement deployment in affected regions [38]. **(2) Spacecraft Collisions.** Starlink satellites are involved in about 1,600 close encounters per week, comprising over half of all such incidents in orbit as of 2021 [43]. This number is anticipated to increase to 90% as the network expands [29]. Collisions can cause direct hardware damage and generate space debris, posing cascading risks to satellite constellations. **(3) Military Attacks.** LEO satellites, including Starlink, are targeted in military conflicts for their critical communication capabilities. These satellites are vulnerable to anti-satellite missile tests aimed at disrupting enemy networks, impacting communication and surveillance systems globally [33, 40, 41]. **(4) Extreme Space Weather.** In February 2022, Starlink launched a batch of satellites followed by a subsequent geomagnetic storm struck, resulting in the loss of 40 Starlink internet satellites [22]. Solar storms are also causing satellites to descend from orbit [17]. Satellites exposed in space lack the protective shield of the atmosphere and Earth’s magnetic field, making them vulnerable to direct threats from radiation and magnetic storms. Shortly, as satellite deployment grows and networks transition to inter-satellite links (ISLs), the probability of satellite-related risks will rise accordingly. The reliability of satellite networks is pivotal for both everyday internet users and essential internet services, impacting economic activities and international relations. Hence, conducting thorough risk assessments for the entire satellite network is crucial and urgent for its future development.

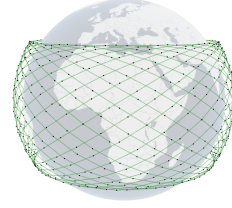


Figure 1: Illustration of LEO Satellite Constellation Shell, showcasing Shell 3 of Kuiper.

Currently, research on network resilience for next-generation satellite networks using Inter-Satellite Links (ISLs) is limited [17, 23, 42]. Given that existing satellite constellations are in a transitional phase combining bent-pipe and ISL architectures, assessing the impact of satellite damage on ISL-based networks is impractical with real-world setups. Moreover, since physically destroying or disabling LEO satellites is not feasible, simulations are necessary for analyzing ISL-based satellite networks. To our knowledge, Our study is the first to evaluate network resilience and performance under potential physical damage to ISL-based constellations. Ultimately, we aim to develop a constellation simulation that mirrors real-world conditions, integrating both bent-pipe and ISL architectures for comprehensive attack and connectivity analyses, and to create full-layer network traffic simulations for multi-shell constellations to reflect realistic satellite network scenarios.

This paper presents our early-stage research on the resilience performance of a single-layer ISL-based satellite network under possible satellites failures. We selected the top 100 most populous cities on Earth as traffic sources for analyzing the ISL-based satellite network. Our findings reveal several insights that are expected to hold across a broad range of traffic matrices.

2 BACKGROUND & RELATED WORK

This section provides an preliminary knowledge of LEO satellite networks and the recent work on LEO Constellation performance measurement.

LEO Satellite Constellation. LEO satellite constellation may include thousands of satellites distributed across one or more orbital shells, with each shell consisting of specific satellite orbits. To describe the shells of an LEO constellation, several parameters are required: (a) The *altitude* specifies the height of the satellite orbit above sea level, affecting the satellite’s velocity and coverage area; (b) The *number of orbits*; (c) The *number of satellites per orbit*, along with the *number of orbits*, determines the overall density of the constellation. Satellites on the same orbit are typically evenly distributed; (d) The *orbital inclination* is the angle between the orbital plane and the equatorial plane. Graphically, the inclination represents the maximum latitude that the vertical projection

of a satellite on the ground can reach. Typically, orbits within a shell have the same *altitude*, *number of satellites per orbit*, and *inclination*, defining the shell’s parameters. Orbits are evenly spaced within a shell, with their intersection points on the equatorial plane uniformly distributed around a circle.

LEO Constellation Performance Measurement. Recent works [15, 18, 21, 24] focus on measuring existing LEO constellations. However, due to the difficulty of deploying large-scale real-world measurements of LEO constellations, simulation based research become a viable method to gain high-level insights into overall performance. Previously, Basak et al.[8] suggested using lower budgets to design better constellations. Zhang et al.[42] investigate the impact of different constellation parameters on network performance through simulations. Bhosale et al. [3] characterize route variability in LEO satellite networks. However, prior work [4, 8, 27] only explore the performance of constellations over a limited range of parameter settings and did not examine the potential impact of satellite network failures. To our knowledge, no work has focused on the network resilience of the LEO constellation. This work aims to fill this gap.

3 EXPERIMENT DESIGN

This experiment is to reveal the workload distribution of mature commercial LEO satellites and evaluate network resilience under critical satellite disconnection.

(i) **Geographic Features.** This section investigates the correlation between the number of ground stations connected by satellites and the network connections facilitated by those satellites, revealing the geographic distribution of network workload.

(ii) **Critical Satellites Disconnection.** This section simulates the impact of critical satellite disconnection on constellation performance, characterized by RTT-related metrics. By removing several heavily used satellites, we visualize the network’s load transfer and analyze performance changes, highlighting key impacts on the constellation.

We choose Starlink’s first Shell (S1) and Kuiper’s third Shell (K3) for analysis, using the 100 most populous cities [28] as ground stations. Starlink’s S1 [34] consists of 1,584 satellites (72 orbits, each with 22 satellites) at an altitude of 550 km and an inclination of 53°. Kuiper’s K3 [1] comprises 784 satellites (28 orbits, each with 28 satellites) at an altitude of 590 km and an inclination of 33°.

For a precise analysis of the LEO satellite network resilience, we employ a simulator based on Hypatia [19], which models the satellites’ relative motion and interactions with ground stations. The simulation uses a 200-second interval (sufficient to cover the period of one satellite replaces neighbor’s previous position) with a 1-second time step. At each time step, we assume that all city pairs have traffic and ideal

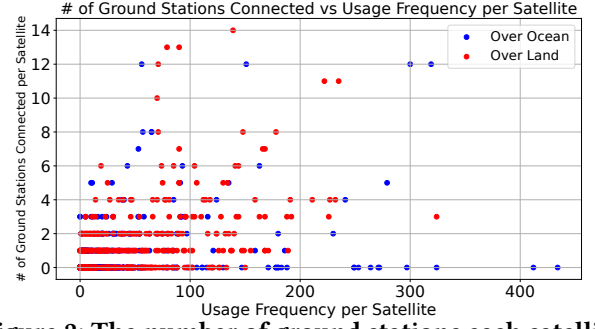


Figure 2: The number of ground stations each satellite connects to and the overall usage frequency of each satellite in the shortest paths between all ground station pairs at time-stamp 0.

transmission conditions with instantaneous packet transfer with no packet loss. Transmission routes are optimized based on the shortest geographical path using Dijkstra’s algorithm. This approach allows us to monitor the usage frequency of each satellite in the paths between ground stations and evaluate the Round Trip Time (RTT) and Geodesic Slowdown for city pairs’ traffic. By generating static network models at successive time steps, we analyze the resilience performance of the entire LEO satellite network.

4 EVALUATION

4.1 Geographic Features

This section examines satellite usage frequency to assess satellite network performance. Satellite usage frequency reflects the load on each satellite by counting how many city pairs it serves as part of the shortest communication path during a single time step. It gives a realistic assessment of how each satellite supports city-to-city traffic.

4.1.1 Connectivity vs Usage. To explore the relationship between the number of ground stations each satellite connects to and its usage frequency among city pairings’ traffic, we use shell 1 of Starlink as an example. Connectivity refers to the total number of ground stations a satellite can actively connect to within its operational range. While this number may exceed actual connections due to antenna capabilities, it represents the satellite’s potential coverage.

We categorize satellites by geographic location: Over Ocean (with few or no ground station connections) and Over Land (linked to many ground stations). In Fig. 2, we note that most satellites have fewer than 6 ground station connections and a usage frequency below 100. Interestingly, there is no clear positive correlation between the number of ground station connections and usage frequency. Many satellites in areas with dense ground stations (> 13) are underutilized (< 200 uses), while frequently used satellites (> 400 uses) are often over oceans without ground station connections. To further

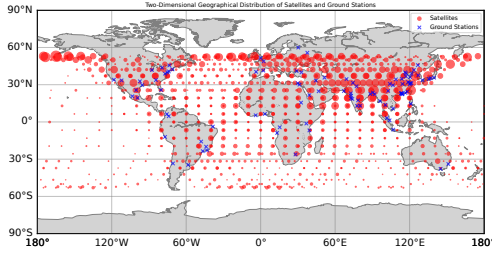


Figure 3: 2D geographical distribution of S1 satellites and ground stations at times-tamp 0. The size of red circle reflects the usage frequency of the corresponding satellite – larger dots representing higher usage rates.

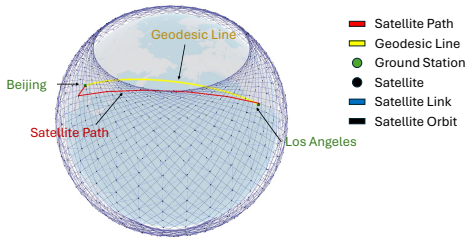


Figure 4: Shortest ISLs Link vs. Geodesic Line Between Beijing and Los Angeles.

investigate this, we visualize and analyze the geographical features of satellite usage in § 4.1.2.

4.1.2 Geographical Features vs Usage. Fig. 3 illustrates the global distribution of ground stations and the usage frequency of S1’s satellites at a specific timestamp. Satellites with high traffic are primarily in the northern hemisphere, correlating with the higher density of ground stations. Ground station density significantly influences the Over Land Satellites’ utilization, particularly in densely populated regions like Asia, Europe, and the North American coastline. The Over Ocean satellites above the northernmost Pacific and Atlantic Oceans show high utilization rates, highlighting their importance in transcontinental communications, as indicated by the extreme scatter points in Fig. 2.

This phenomenon can be explained by the geodesic lines’ shortest paths from numerous ground stations in different northern hemisphere continental plates, as illustrated in Fig. 4, which must pass over higher latitude oceans. However, with S1’s inclination limited to 53°, these paths are forced to converge on the satellites located at the highest latitudes of the constellation - they lie near the shortest paths.

Geographic Features Key Takeaways:

- Many satellites are underutilized or unused.
- Ground station density influences the Over Land Satellites’ utilization, particularly in densely populated regions like Asia, Europe, and the North American coastline.

- The Over Ocean satellites above the northernmost Pacific and Atlantic Oceans show high utilization rates because the constellation’s inclination forces most of the over ocean ground stations’ traffic through the constellation’s edge satellites.

4.2 Critical Satellites Disconnection

To identify satellites crucial for maintaining high network performance, we conduct an in-depth investigation. We hypothesize that satellites with high usage frequency are critical to the network. Thus, Our experiment examines changes in utilization and performance metrics after removing the top-n most utilized satellites to simulate critical satellite disconnections. The loss of these hotspot satellites, which significantly diminishes constellation performance, simulates the effects of military attacks [20]. Once the top-N most used satellites fail, a shortest path algorithm is employed via the remaining operational satellites.

We examine two constellations: Shell 1 (S1) of Starlink and Shell 3 of Kuiper (K3). S1 features broader coverage due to its larger inclination and greater number of satellites, while K3 has a smaller inclination and fewer satellites, concentrated at lower latitudes. Due to the differing number of satellites in each constellation, we conduct distinct disconnection investigations. For S1, we simulate the disconnection of 100, 200, and more satellites. For K3, we simulate the disconnection of 50, 100, and 150 satellites.

4.2.1 Changes in Satellite Usage. As shown in Fig. 5(a), 5(b) and 5(c), the utilization rate of satellites near the removed satellites significantly increases, compensating for the connections originally handled by the removed satellites. Interestingly, many non-adjacent satellites, such as those over northern Africa, also experience a marked increase in utilization. This results from substantial changes in communication paths following the removal of critical satellites. This observation supports the conclusion discussed in § 4.1: some satellites not directly connected to ground stations are utilized because they lie near the shortest paths between selected cities. Removing these satellites forces many ground stations to establish entirely different communication paths.

4.2.2 RTT Analysis. This work focuses on non-congestive round-trip time (RTT), which refers to the average round-trip time recorded under ideal circumstances, without taking into account network congestion. This establishes a standard for assessing latency. Prior research have also taken into account comparable optimal conditions. [19, 42] We concentrated on two main metrics: (a) *Max. RTT* is the maximum Round Trip Time at the speed of light observed between the sender and receiver during a specified period. (b) *Geodesic Slowdown* measures the slowdown relative to the optimal path, defined

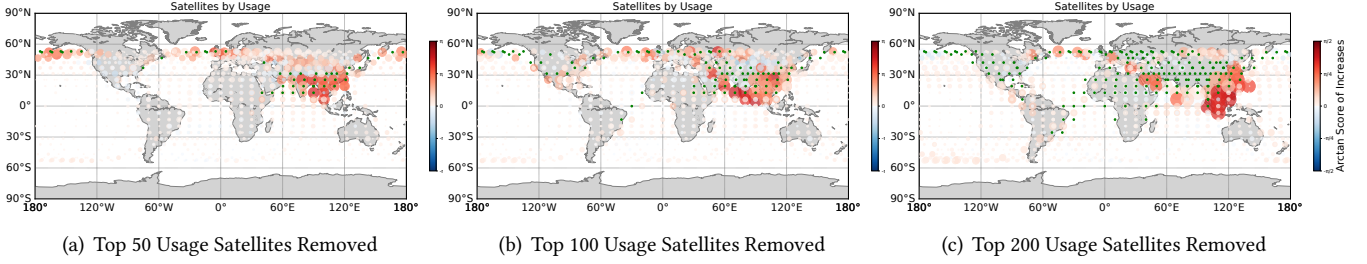


Figure 5: Maps of Satellite Utilization after removing Critical Satellites. In these maps, satellites are represented by colored dots. The size of the dot indicates the satellite’s usage, and the color of the dot represents the rate of change in usage before and after the satellite removal. Deeper red indicates a greater increase in usage, while deeper blue indicates a higher decrease in usage. Removed satellites are indicated by green points.

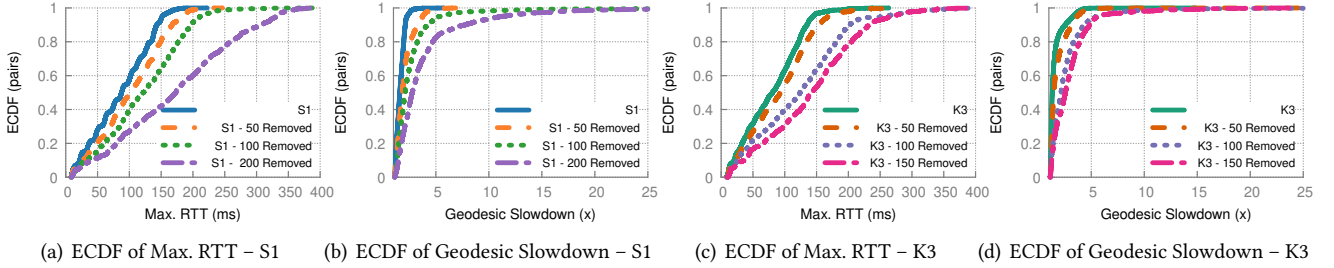


Figure 6: Empirical Cumulative Distribution Functions (ECDF) of Max. RTT and Geodesic Slowdown of S1 and K3.

as $S_g = \frac{\max.RTT}{ideal.RTT}$, $ideal.RTT$ is the RTT for direct data transmission between two endpoints over the shortest Euclidean distance on the earth’s surface at the speed of light.

As shown in Fig. 6, our results indicate that both constellations’ Max. RTT and Geodesic Slowdown decreased when the critical satellite disconnection number increased. For example, 60% of the traffic experience an increase in delay from 100ms to 200ms after the removal of 200 critical satellites in S1. With the increase in the number of disconnected satellites, the ECDF of performance metrics shows a more pronounced long-tail effect. The RTT of the highest delay connections increases significantly, indicating that the removal of satellites can be detrimental to network connectivity.

Critical Satellites Disconnection Takeaways:

- The removal of satellites will significantly increase the utilization rate of surrounding satellites. This implies that in real-world scenarios, the surrounding satellites are likely to face instantaneous high loads to compensate for the disconnected satellites.
- Some satellites far from the removed satellites become new high-load satellites due to changes in shortest paths.
- Critical Satellite Disconnection significantly increase the overall network Max. RTT and Geodesic slowdown, detrimental to network performance.

5 DISCUSSION

In this section, we identify avenues for future work.

Conduct Dynamic Analyses In this work, we analyzed the impact of disconnecting a large number of critical satellites on overall network performance through simulation experiments, which were primarily statistical and static. However, real-world scenarios typically involve the disconnection of a few satellites due to failures or attacks, affecting communication between specific nodes, such as key cities. Therefore, conducting dynamic analyses is crucial to understanding the security and robustness of satellite network communication in these contexts.

More Realistic Simulation We will also take into account more realistic factors to replace the idealized assumptions. These factors include satellite network utilization distribution based on real data, more realistic inter-satellite communication implementation, and considering the impact of inter-satellite network rerouting, among others. We also plan to investigate the possible advantages of including multiple mesh networks into satellite constellations.

6 CONCLUSION

In this work, we utilize Starlink Shell 1 and Kuiper Shell 3 to analyze resilience of the ISLs-based LEO satellite network. Our analysis reveals that many satellites, especially those positioned over oceans, are not actively used and that ground station density significantly influences satellite utilization, particularly in densely populated regions like Asia, Europe, and the North American coastline. Additionally, Over Ocean satellites above the northernmost Pacific and Atlantic Oceans

exhibit high usage due to the constellation's inclination, which directs most over-ocean traffic through edge satellites. Lastly, Removing critical satellites significantly increased the utilization of nearby and distant satellites, worsening network performance. These findings emphasize the need for robust satellite distribution and resilience planning to maintain optimal performance. Further research on dynamic network adaptation and redundancy strategies is necessary to mitigate the effects of satellite disconnections.

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